# Thermodynamic Consistency, Structural Entropy, and Rosenfeld Diffusion Scaling in Multicomponent Electrolytes

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**Abstract**

This report develops the classical entropy reference, multicomponent pair entropy, and internal HNC free-energy/virial consistency tests, retaining historical numerical benchmarks as a separate theoretical record. It then incorporates the completed HNC-guided replicated MD study of eight mixtures with fixed divalent fractions, $N = 4000$, three replicas, and 19 densities per replica. A fixed LJ Hamiltonian, configurational pressure reference, and Debye–Hückel endpoint provide excess entropy; a common system COM frame provides transport. High-entropy solvent designs increase probe diffusion by 18–20% relative to the baselines. At matched charged composition and mean bare LJ amplitude, diversification raises excess entropy by approximately $0.103k_B$ and gives a smaller, provisional probe increase of approximately 5%. Label-only splitting leaves diffusion identical, and monovalent addition alone gives no resolved speedup. Positive descriptive Rosenfeld-like trends now appear for both mean and probe mobility, but short fitting-window sensitivity, baseline charge-sign asymmetry, and finite-size uncertainty preclude a universal species law. The former two-state $B \simeq 2.44$ transport interpretation is superseded. A corrected classical 2PT analysis passes internal and label checks but retains composition-dependent entropy biases, supporting spectral diagnostics while leaving thermodynamic integration as the primary entropy estimate.

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# 1 Introduction

In the statistical mechanics of dense liquids and ionic solutions, integral equation theory based on the Ornstein–Zernike (OZ) equation provides an efficient, non-perturbative alternative to computer simulations. For a multicomponent mixture of $m$ species with total number density $\rho = N/V$, mole fractions $x_i = N_i/N$, and partial number densities $\rho_i = \rho x_i$, the structural state is characterized by the matrix of total correlation functions $h_{ij}(r) = g_{ij}(r) - 1$ and direct correlation functions $c_{ij}(r)$.

The Hypernetted Chain (HNC) closure sets the bridge function identically to zero ($B_{ij}(r) \equiv 0$):

$$g_{ij}(r) = \exp\left(-\beta v_{ij}(r) + \gamma_{ij}(r)\right), \tag{1}$$

where $\gamma_{ij}(r) = h_{ij}(r) - c_{ij}(r)$ is the indirect correlation function, $\beta = 1/(k_B T)$, and $v_{ij}(r)$ is the pair potential containing short-range (Lennard-Jones) and long-range (Coulomb) interactions:

$$v_{ij}(r) = 4\epsilon_{ij}\left[\left(\frac{\sigma_{ij}}{r}\right)^{12} - \left(\frac{\sigma_{ij}}{r}\right)^{6}\right] + \frac{q_i q_j e^2}{4\pi\varepsilon_0\varepsilon_r r}. \tag{2}$$

For Coulombic systems, the long-range tail is screened and renormalized using Allnatt's formulation [1], preventing Fourier divergences at $k \to 0$.

Once the structural correlations are converged, macroscopic thermodynamic quantities can be obtained via several mechanical and statistical routes:

1. **Virial Route**: Pressure $P_{\text{vir}}$ computed from the average virial of intermolecular forces:

$$Z_{\text{vir}} = \frac{P_{\text{vir}}}{\rho k_B T} = 1 - \frac{2\pi\rho}{3}\sum_{i,j} x_i x_j \int_0^\infty r^3 \frac{dv_{ij}(r)}{dr} g_{ij}(r)\, dr. \tag{3}$$

2. **Energy Route**: Excess internal energy $u_{\text{ex}}/k_B T$ computed from the pair potential weighted by $g_{ij}(r)$.

3. **Generating Functional Route**: Closed-form evaluation of the excess Helmholtz free energy $a_{\text{ex}}^{\text{HNC}} = A_{\text{ex}}/(N k_B T)$ and excess chemical potentials $\mu_i^{\text{ex}}/k_B T$ from $k$-space and $r$-space integrals [2–4].

4. **Compressibility Route**: Isothermal compressibility $\chi_T$ evaluated from the $k \to 0$ limit of the direct correlation functions.

In exact statistical mechanics, all routes yield identical thermodynamic functions. In approximate closures, however, inconsistencies arise. A central focus of this report is demonstrating how the virial route and the HNC free energy functional relate, establishing their internal consistency through numerical differentiation and path integration.

# 2 Classical vs. Quantum Entropy: Why Classical Entropy Can Be Negative

When computing the total entropy per particle $s = s_{\text{id}} + s_{\text{ex}}$ in reduced units ($\sigma = 1, \epsilon = 1, k_B = 1$), one often observes negative total entropy at low temperatures or high ionic strengths. This section explains why this is an entirely physical and expected property of continuous classical statistical mechanics.

### 2.1 Phase-Space Divergence at Zero Temperature

In quantum statistical mechanics, the entropy is defined via the discrete density matrix $\hat{\rho}$ as $S = -k_B \mathrm{Tr}(\hat{\rho} \ln \hat{\rho})$. By the Third Law of Thermodynamics (Nernst Heat Theorem), as $T \to 0$, the system settles into its non-degenerate ground state:

$$\lim_{T \to 0} S_{\text{quantum}} = 0 \quad (\text{or } S \geq 0 \text{ strictly}). \tag{4}$$

In classical mechanics, microstates occupy a continuous phase space $\Gamma = (\mathbf{r}^N, \mathbf{p}^N)$. The classical Gibbs entropy is defined as an integral over the continuous probability density $f(\Gamma)$:

$$S_{\text{classical}} = -k_B \int f(\Gamma) \ln[h^{3N} f(\Gamma)] \, d\Gamma. \tag{5}$$

Because $f(\Gamma)$ has physical dimensions of $(\text{action})^{-3N}$, the argument of the logarithm involves the continuous probability density. For an ideal gas of particles with mass $m$ at temperature $T$, the momentum distribution is Gaussian with variance $\langle p^2 \rangle = m k_B T$. As $T \to 0$, the momentum distribution narrows toward a Dirac delta function $\delta^3(\mathbf{p})$, causing the momentum-space volume to shrink to zero. Consequently:

$$\lim_{T \to 0} S_{\text{classical}} = -\infty. \tag{6}$$

A negative value of classical entropy simply reflects the fact that the continuous distribution is more localized in phase space than an arbitrary reference volume of size $h^{3N}$.

### 2.2 Reduced Units and the Thermal de Broglie Wavelength

The classical ideal-gas entropy for a multicomponent mixture of non-interacting particles is:

$$\frac{s_{\text{id}}}{k_B} = \frac{5}{2} - \sum_{i=1}^{m} x_i \ln\left(\rho_i \Lambda_i^3\right), \quad \Lambda_i = \frac{h}{\sqrt{2\pi m_i k_B T}}. \tag{7}$$

Factoring out an arbitrary reference atomic length scale $\sigma$ (such as the Lennard-Jones diameter $\sigma \equiv 1$):

$$\frac{s_{\text{id}}}{k_B} = \underbrace{\left[\frac{5}{2} - \ln(\rho\sigma^3) - \sum_{i=1}^{m} x_i \ln x_i\right]}_{\text{Reduced-Unit Baseline } (s_{\text{id}})_{\text{reduced}}} - \underbrace{3 \sum_{i=1}^{m} x_i \ln\left(\frac{\Lambda_i}{\sigma}\right)}_{\text{Quantum Phase-Space Offset } \Delta s_{\text{quantum}}}. \tag{8}$$

In standard simulation and integral equation codes, calculations are performed in reduced units where $\sigma = 1$, $\epsilon = 1$, and masses are omitted. The program evaluates only the bracketed term in Eq. (8), implicitly setting $\Lambda_i = \sigma = 1$.

For physical fluids (such as argon, water, or aqueous electrolytes) at room temperature:

$$\sigma \approx 3.0\text{–}4.0 \text{ Å}, \quad \Lambda \approx 0.15\text{–}0.25 \text{ Å} \implies \frac{\Lambda}{\sigma} \approx 0.04 \ll 1. \tag{9}$$

The omitted quantum phase-space term contributes:

$$\Delta s_{\text{quantum}} = -3 \ln\left(\frac{\Lambda}{\sigma}\right) \approx -3 \ln(0.04) \approx +9.66 \, k_B. \tag{10}$$

Restoring this phase-space offset shifts the absolute entropy upward by approximately 8–12$k_B$ per particle for the illustrative mapping. It fixes the omitted normalization but does not extend the classical approximation into the quantum-degenerate regime. A physical absolute entropy requires the masses and length scale; excess entropy relative to the corresponding ideal mixture is independent of this common normalization.

# 3 The Multicomponent Two-Body Pair Entropy $s^{(2)}$

## 3.1 Green–Wallace Expansion

The thermodynamic excess entropy $s_{\text{ex}} = (u_{\text{ex}} - a_{\text{ex}})/k_BT$ measures the reduction in accessible configurational space due to interparticle interactions. It can be expanded in an infinite series of irreducible multiparticle spatial correlations (Green, Kirkwood, Nettleton, and Wallace):

$$s_{\text{ex}} = s^{(2)} + s^{(3)} + s^{(4)} + \dots \tag{11}$$

In dense simple fluids and ionic solutions, the two-body contribution $s^{(2)}$ accounts for the majority (75%–95%) of the total excess entropy.

## 3.2 Exact Multicomponent Formulation

For an isotropic fluid mixture of $m$ species with mole fractions $x_i$, the exact two-body structural entropy per particle is:

$$\frac{s^{(2)}}{Nk_B} = -2\pi\rho \sum_{i=1}^{m} \sum_{j \le i} (2 - \delta_{ij})\, x_i x_j \int_0^\infty [g_{ij}(r) \ln g_{ij}(r) - h_{ij}(r)]\, r^2 dr. \tag{12}$$

## 3.3 Mathematical Properties

1. **Negative Definiteness**: The function $f(g) = g \ln g - g + 1$ is strictly convex for all $g \ge 0$, with a unique minimum at $g = 1$ where $f(1) = 0$. Because $f(g) \ge 0$ everywhere:

$$\frac{s^{(2)}}{Nk_B} \le 0 \quad \text{strictly.} \tag{13}$$

2. **Core Behavior**: Inside the repulsive core where particles cannot overlap ($g_{ij}(r) \to 0$):

$$\lim_{g \to 0} g \ln g = 0 \implies [g \ln g - h] = -(-1) = 1. \tag{14}$$

   Integrating over the excluded volume from $r = 0$ to $r = R_{\text{core}}$ yields an excluded-volume entropic penalty of $-\frac{2\pi}{3}\rho R_{\text{core}}^3 x_i x_j$.

3. **Asymptotic Decay**: For large $r$, $h(r) \to 0$. Expanding $f(g)$:

$$(1 + h)\ln(1 + h) - h = \frac{1}{2}h^2 - \frac{1}{6}h^3 + \mathcal{O}(h^4). \tag{15}$$

   Because electrostatic interactions in electrolytes are exponentially screened (Debye screening), $h(r)$ decays rapidly, ensuring absolute convergence of the integral.

# 4 Thermodynamic Consistency of the HNC Generating Functional

## 4.1 The HNC Functional and the Euler Relation

In the HNC approximation, the direct correlation function $c_{ij}(r)$ and chemical potentials $\mu_i^{\text{ex}}$ are generated by a closed-form functional $\mathcal{F}_{\text{ex}}^{\text{HNC}}[\{\rho_i\}]$ [2,3]. By Euler's theorem for homogeneous functions of first degree in $(V, \{N_i\})$:

$$A_{\text{ex}} = -P_{\text{ex}}V + \sum_{i=1}^{m} \mu_i^{\text{ex}} N_i. \tag{16}$$

Dividing by $Nk_BT$ and rearranging yields the local Euler relation for the compressibility factor:

$$Z(a) - 1 = \sum_{i=1}^{m} x_i \frac{\mu_i^{\text{ex}}}{k_B T} - a_{\text{ex}}^{\text{HNC}}. \tag{17}$$

Because the HNC closure is stationary with respect to variations of $h_{ij}$ and $c_{ij}$, the chemical potentials and excess free energy satisfy this relation identically at every state point.

### 4.2 Differentiation of Free Energy vs. Virial Equation of State

From fundamental thermodynamics, the pressure is the negative volume derivative of the Helmholtz free energy at fixed temperature and composition:

$$P = -\left(\frac{\partial A}{\partial V}\right)_{T,\{N_i\}} = -\left(\frac{\partial (N k_B T a)}{\partial (N/\rho)}\right) = \rho^2 k_B T \left(\frac{\partial a}{\partial \rho}\right)_{T,\{x_i\}}. \tag{18}$$

Subtracting the ideal gas contribution $P_{\text{id}} = \rho k_B T$:

$$Z_{\text{diff}} - 1 = \frac{P_{\text{ex}}}{\rho k_B T} = \rho \left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_{T,\{x_i\}}. \tag{19}$$

Equivalently, differentiating the free energy per unit volume $\mathcal{F}_{\text{ex}}/V = \rho a_{\text{ex}}$ with respect to total density yields the composition-averaged chemical potential:

$$\frac{\partial (\rho a_{\text{ex}})}{\partial \rho} = a_{\text{ex}} + \rho \frac{\partial a_{\text{ex}}}{\partial \rho} = \sum_{i=1}^{m} x_i \frac{\mu_i^{\text{ex}}}{k_B T}. \tag{20}$$

Comparing this with Eq. (17) demonstrates that the local Euler compressibility factor $Z(a)$ and the derivative compressibility factor $Z_{\text{diff}}$ are mathematically equivalent if and only if:

$$\left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right) = \frac{Z - 1}{\rho}. \tag{21}$$

### 4.3 Numerical Stencils for Differentiation

To evaluate Eq. (19) from discrete density points with uniform grid spacing $\Delta\rho$, we employ symmetric 3-point and 5-point finite difference stencils:

$$\left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_i^{(3\text{pt})} = \frac{a_{i+1} - a_{i-1}}{2\Delta\rho} + \mathcal{O}(\Delta\rho^2), \tag{22}$$

$$\left(\frac{\partial a_{\text{ex}}}{\partial \rho}\right)_i^{(5\text{pt})} = \frac{-a_{i+2} + 8a_{i+1} - 8a_{i-1} + a_{i-2}}{12\Delta\rho} + \mathcal{O}(\Delta\rho^4). \tag{23}$$

## 5 Restricted Primitive Model Benchmark

To verify that the HNC functional generates the virial equation of state upon differentiation, we first examine the classical Restricted Primitive Model (RPM): equal-diameter hard spheres ($\sigma = 1.0$) with symmetric $\pm 1$ charges at $\beta^* = 1.00$.

Table 1 presents the comparison between $Z_{\text{virial}}$, the Euler route $Z(a)$, and the numerical derivatives evaluated from the converged HNC solution.

The 5-point numerical derivative of the HNC free energy reproduces the virial compressibility factor to four significant digits (relative error $\leq 0.03\%$), confirming the exact internal consistency of the HNC formulation for hard-core reference systems.

Table 1: Numerical differentiation of the HNC excess free energy for the RPM hard-sphere electrolyte at $\beta^* = 1.00$ ($\Delta\rho = 0.10$).

| $\rho^*$ | $Z_{\text{virial}}$ | $Z(a)$ (Euler) | $Z_{3\text{pt}}$ | Error (3pt) | $Z_{5\text{pt}}$ | Error (5pt) |
|---|---|---|---|---|---|---|
| 0.20 | 1.4824 | 1.4824 | 1.4811 | $-0.0013$ ($-0.08\%$) | — | — |
| 0.30 | 1.9212 | 1.9212 | 1.9273 | $+0.0061$ ($+0.32\%$) | **1.9218** | $+$**0.0006** ($+$**0.03**%) |
| 0.40 | 2.5403 | 2.5403 | 2.5542 | $+0.0139$ ($+0.55\%$) | **2.5397** | $-$**0.0006** ($-$**0.02**%) |
| 0.50 | 3.4224 | 3.4225 | 3.4485 | $+0.0261$ ($+0.76\%$) | — | — |

# 6 Resolution of the Core-Radius Discrepancy in Continuous Potentials

## 6.1 The Origin of the Discrepancy

When adapting the legacy solver from hard spheres (RPM) to Lennard-Jones fluids in 'HNCR-nCLJ.f90', the hard core boundary was adjusted:

$$ncore(j) = \text{nint}\left(\frac{0.6\,\sigma_j}{\Delta r}\right) \implies R_{\text{core}}(j) = 0.6\,\sigma_j. \tag{24}$$

Because Lennard-Jones particles have soft repulsive cores, they penetrate inside the nominal diameter $\sigma_j = 1.0$ down to $r \approx 0.6\sigma_j$ at high temperatures. All radial integration loops for the correlation functions split at $R_{\text{core}}(j)$:

- Backward Simpson quadrature from $R_{\text{core}}$ down to 0 for the inner cavity direct correlation.
- Forward Simpson quadrature from $R_{\text{core}}$ to $R_{\max}$ for the fluid correlation functions.

However, in the original implementation of the thermodynamic subroutine:

1. The hard-core cavity free energy term $a_{\text{SR}}$ and chemical potential shifts:

$$a_{\text{SR}} = \frac{2\pi}{3\rho}\sum_{i,j}\rho_i\rho_j\sigma_{ij}^3 \tag{25}$$

   were still evaluated using the nominal diameter $\sigma_{ij} = 1.0$ rather than the actual integration cutoff boundary $R_{\text{core}}(ij) = 0.6\,\sigma_{ij}$.

2. The electrostatic surface charging term in $ekt(j)$ was evaluated using $\pi\gamma_j\sigma_j^2$ rather than $\pi\gamma_j R_{\text{core}}(j)^2$.

## 6.2 Analytical Error Quantification

Because the inner integration was performed up to $R_{\text{core}} = 0.6\sigma$, but the cavity volume subtraction assumed a core radius of $1.0\sigma$, an artificial excess volume was subtracted from the free energy:

$$\Delta a_{\text{ex}}(\rho) = \frac{2\pi}{3}\left[\sigma^3 - R_{\text{core}}^3\right]\rho = \frac{2\pi}{3}\left[1.0^3 - 0.6^3\right]\rho = \mathbf{1.6420}\,\rho. \tag{26}$$

While this spurious $+1.6420\,\rho$ cancelled algebraically in the local Euler relation $Z(a) = 1+\mu^{\text{ex}}-a_{\text{ex}}$, its density derivative shifted the derivative compressibility factor by:

$$\Delta Z_{\text{diff}} = \rho\frac{\partial(\Delta a_{\text{ex}})}{\partial\rho} = \mathbf{1.6420}\,\rho. \tag{27}$$

At $\rho^* = 0.40$, this caused an error of $+1.6420 \times 0.40 = +0.6568$, precisely explaining why earlier numerical derivatives on the uncorrected code differed from $Z_{\text{virial}}$.

### 6.3 Code Correction

In `src/thermo_mod.f90`, the core radius reference in the cavity and electrostatic surface integrals was updated from $\sigma_j$ to $R(ncore(j))$:

$$ekt(j) = \frac{2\pi}{3} esum\, \Delta r + \pi\gamma_j R(ncore(j))^2, \tag{28}$$

$$sm1 = \sum_{i,j} \rho_i \rho_j R(ncore(npos(i,j)))^3. \tag{29}$$

This completely eliminates the artificial slope, restoring exact consistency between the free energy functional and the virial equation of state.

## 7 Historical HNC Benchmarks for Different Compositions

### 7.1 System Specifications

The following historical HNC benchmarks concern two compositions distinct from the replicated MD states defined later. Their labels 3C and 5C apply only within these benchmark sections; they must not be identified with B3 and B5. We consider the isotherm of these two fluids along the supercritical isotherm $T^* = 2.00$ ($\beta^* = 0.50$, electrostatic coupling $\gamma = 1.0$):

- **System 3C**: Neutral solvent ($x_1 = 0.80$, $q_1 = 0$) + symmetric 2:2 salt ($x_2 = 0.10, q_2 = +2$; $x_3 = 0.10, q_3 = -2$). Total ionic strength $I = \frac{1}{2}\sum x_i q_i^2 = 0.40$.
- **System 5C**: Neutral solvent ($x_1 = 0.80$, $q_1 = 0$) + mixed 1:1 and 2:2 salts ($x_2 = 0.05, q_2 = +1$; $x_3 = 0.04, q_3 = +2$; $x_4 = 0.09, q_4 = -1$; $x_5 = 0.02, q_5 = -2$). Total ionic strength $I = 0.19$.

Both systems are matched to identical Lennard-Jones parameters ($\sigma = 1.0, \epsilon = 1.0, r_{\text{cut}} = 2.5$) and identical packing fraction ($\eta \approx 0.3665$ at $\rho^* = 0.70$).

### 7.2 Differentiation Consistency Benchmark

Tables 2 and 3 display the results of numerical differentiation of $a_{\text{ex}}^{\text{HNC}}(\rho)$ compared with $Z_{\text{virial}}$ across the entire density range $\rho^* \in [0.05, 0.70]$ with uniform spacing $\Delta\rho = 0.05$.

Table 2: Numerical differentiation of HNC free energy for System 3C (Solvent + 2:2 Salt).

| $\rho^*$ | $Z_{\text{virial}}$ | $Z(a)$ (Euler) | $Z_{3\text{pt}}$ | Error (3pt) | $Z_{5\text{pt}}$ | Error (5pt) | Rel. Error |
|---|---|---|---|---|---|---|---|
| 0.10 | 0.9188 | 0.9192 | 0.9175 | −0.0013 | — | — | — |
| 0.15 | 0.9058 | 0.9064 | 0.9055 | −0.0003 | **0.9061** | +0.0003 | **0.03%** |
| 0.20 | 0.9078 | 0.9087 | 0.9082 | +0.0004 | **0.9081** | +0.0003 | **0.04%** |
| 0.25 | 0.9276 | 0.9288 | 0.9285 | +0.0009 | **0.9278** | +0.0002 | **0.03%** |
| 0.30 | 0.9694 | 0.9710 | 0.9709 | +0.0015 | **0.9695** | +0.0001 | **0.02%** |
| 0.35 | 1.0390 | 1.0410 | 1.0416 | +0.0027 | **1.0397** | +0.0007 | **0.06%** |
| 0.40 | 1.1441 | 1.1466 | 1.1476 | +0.0035 | **1.1447** | +0.0006 | **0.05%** |
| 0.45 | 1.2941 | 1.2972 | 1.2983 | +0.0042 | **1.2942** | +0.0001 | **0.01%** |
| 0.50 | 1.5005 | 1.5043 | 1.5060 | +0.0055 | **1.5007** | +0.0002 | **0.01%** |
| 0.55 | 1.7766 | 1.7814 | 1.7837 | +0.0071 | **1.7771** | +0.0005 | **0.03%** |
| 0.60 | 2.1375 | 2.1433 | 2.1466 | +0.0091 | **2.1385** | +0.0010 | **0.05%** |
| 0.65 | 2.5999 | 2.6070 | 2.6107 | +0.0108 | — | — | — |

Across the entire fluid density range, the 5-point numerical derivative of the HNC free energy functional reproduces the virial equation of state with 0.01%–0.05% accuracy.

Table 3: Numerical differentiation of HNC free energy for System 5C (Solvent + 1:1 & 2:2 Salts).

| $\rho^*$ | $Z_{\text{virial}}$ | $Z(a)$ (Euler) | $Z_{3\text{pt}}$ | Error (3pt) | $Z_{5\text{pt}}$ | Error (5pt) | Rel. Error |
|---|---|---|---|---|---|---|---|
| 0.10 | 0.9336 | 0.9339 | 0.9333 | −0.0003 | — | — | — |
| 0.15 | 0.9216 | 0.9222 | 0.9219 | +0.0003 | **0.9219** | +0.0003 | **0.03%** |
| 0.20 | 0.9244 | 0.9253 | 0.9250 | +0.0006 | **0.9246** | +0.0002 | **0.02%** |
| 0.25 | 0.9449 | 0.9461 | 0.9460 | +0.0011 | **0.9451** | +0.0002 | **0.02%** |
| 0.30 | 0.9874 | 0.9889 | 0.9892 | +0.0018 | **0.9878** | +0.0004 | **0.04%** |
| 0.35 | 1.0577 | 1.0598 | 1.0602 | +0.0025 | **1.0580** | +0.0003 | **0.03%** |
| 0.40 | 1.1636 | 1.1661 | 1.1672 | +0.0036 | **1.1642** | +0.0006 | **0.05%** |
| 0.45 | 1.3145 | 1.3177 | 1.3191 | +0.0046 | **1.3149** | +0.0004 | **0.03%** |
| 0.50 | 1.5219 | 1.5258 | 1.5275 | +0.0056 | **1.5221** | +0.0002 | **0.01%** |
| 0.55 | 1.7990 | 1.8038 | 1.8063 | +0.0073 | **1.7996** | +0.0006 | **0.03%** |
| 0.60 | 2.1609 | 2.1668 | 2.1700 | +0.0091 | **2.1619** | +0.0010 | **0.05%** |
| 0.65 | 2.6243 | 2.6315 | 2.6347 | +0.0104 | — | — | — |

# 8 Thermodynamic Integration Along Isotherms

## 8.1 Isothermal Path Formulation

Integrating the equation of state along an isotherm from $\rho = 0$ provides an independent route to the excess free energy:

$$\frac{A_{\text{ex}}^{\text{TI}}(\rho)}{Nk_BT} = \int_0^\rho \frac{Z(\rho') - 1}{\rho'}\, d\rho'. \tag{30}$$

For a screened electrolyte, the dilute limit is not a finite second virial coefficient of the unscreened pair potential. Debye–Hückel theory gives $a_{\text{ex}} \propto -\sqrt{\rho}$, so $(Z-1)/\rho$ diverges integrably as $\rho^{-1/2}$. A finite-intercept extrapolation in $\rho$ is therefore inappropriate. The historical integration table below used such an extrapolation and is retained to document the earlier calculation, not as validation of a correct dilute endpoint. The new MD integration instead uses $y = \sqrt{\rho}$ and its finite screened endpoint [Eqs. (34) and (35)].

## 8.2 Comparison of Direct HNC Functional vs. Thermodynamic Integration

Table 4 compares the integrated free energy $a_{\text{ex}}^{\text{TI}}$ with the closed-form HNC functional $a_{\text{ex}}^{\text{HNC}}$.

Table 4: Historical comparison between direct HNC free energy and finite-intercept density integration for the benchmark compositions. The dilute-endpoint treatment is superseded; these offsets are not an integration-accuracy bound for the new study.

| $\rho^*$ | $a_{\text{ex}}^{\text{HNC}}$(3C) | $a_{\text{ex}}^{\text{TI}}$(3C) | $\text{Diff}_{3\text{C}}$ | $a_{\text{ex}}^{\text{HNC}}$(5C) | $a_{\text{ex}}^{\text{TI}}$(5C) | $\text{Diff}_{5\text{C}}$ |
|---|---|---|---|---|---|---|
| 0.05 | −0.0891 | −0.0808 | −0.0083 | −0.0546 | −0.0519 | **−0.0027** |
| 0.10 | −0.1358 | −0.1253 | −0.0105 | −0.0917 | −0.0883 | **−0.0034** |
| 0.20 | −0.1988 | −0.1888 | −0.0100 | −0.1438 | −0.1407 | **−0.0031** |
| 0.30 | −0.2274 | −0.2175 | −0.0099 | −0.1654 | −0.1624 | **−0.0030** |
| 0.40 | −0.2155 | −0.2058 | −0.0097 | −0.1482 | −0.1453 | **−0.0029** |
| 0.50 | −0.1492 | −0.1395 | −0.0097 | −0.0773 | −0.0744 | **−0.0029** |
| 0.60 | −0.0067 | +0.0029 | −0.0096 | +0.0693 | +0.0720 | **−0.0027** |
| 0.70 | **0.2411** | **0.2505** | **−0.0094** | **0.3208** | **0.3235** | **−0.0027** |

At $\rho^* = 0.70$, the historical offsets are $0.0094k_BT$ for 3C and $0.0027k_BT$ for 5C. Their magnitude does not validate the low-density extrapolation: it combines quadrature and the incorrect finite-intercept endpoint. The new calculations do not use this estimator.

# 9 Historical HNC Pair-Entropy Comparison

Table 5 presents the excess entropy $s_{\mathrm{ex}}$, two-body pair entropy $s^{(2)}$, and their differences between the 5-component mixed electrolyte and the 3-component pure 2:2 electrolyte.

Table 5: Entropy comparison between System 3C (pure 2:2) and System 5C (mixed 1:1 + 2:2).

| $\rho^*$ | $s_{\mathrm{ex}}^{3\mathrm{C}}/k_B$ | $s_{\mathrm{ex}}^{5\mathrm{C}}/k_B$ | $\Delta s_{\mathrm{ex}}/k_B$ | $s^{(2),3\mathrm{C}}/k_B$ | $s^{(2),5\mathrm{C}}/k_B$ | $\Delta s^{(2)}/k_B$ | $s^{(2)}/s_{\mathrm{ex}}$ (3C, 5C) | |
|---|---|---|---|---|---|---|---|---|
| 0.01 | −0.0346 | −0.0251 | +0.0095 | −0.0307 | −0.0240 | +0.0067 | 88.7% | 95.6% |
| 0.05 | −0.1286 | −0.1131 | +0.0155 | −0.1185 | −0.1081 | +0.0104 | 92.1% | 95.6% |
| 0.10 | −0.2388 | −0.2211 | +0.0177 | −0.2200 | −0.2086 | +0.0114 | 92.1% | 94.3% |
| 0.20 | −0.4623 | −0.4432 | +0.0191 | −0.4153 | −0.4036 | +0.0117 | 89.8% | 91.1% |
| 0.30 | −0.7030 | −0.6839 | +0.0191 | −0.6109 | −0.5993 | +0.0116 | 86.9% | 87.6% |
| 0.40 | −0.9752 | −0.9562 | +0.0190 | −0.8161 | −0.8047 | +0.0114 | 83.7% | 84.2% |
| 0.50 | −1.2913 | −1.2724 | +0.0189 | −1.0401 | −1.0288 | +0.0113 | 80.5% | 80.9% |
| 0.60 | −1.6606 | −1.6416 | +0.0190 | −1.2928 | −1.2816 | +0.0112 | 77.9% | 78.1% |
| 0.70 | **−2.0886** | **−2.0694** | **+0.0192** | **−1.5853** | **−1.5741** | **+0.0112** | **75.9%** | **76.1%** |

## 9.1 Physical Insights

1. **Enhanced Entropy in the Mixed System**: For these historical benchmark compositions, $\Delta s_{\mathrm{ex}} = s_{\mathrm{ex}}^{5\mathrm{C}} - s_{\mathrm{ex}}^{3\mathrm{C}} = +0.0192\,k_B$ at $\rho^* = 0.70$; the difference is positive at all tabulated densities. The compositions and ionic strengths differ, so this entropy ordering cannot isolate component multiplicity. Association and clustering mechanisms require separate structural and lifetime measurements.

2. **Two-Body Dominance**: The two-body structural entropy $s^{(2)}$ accounts for 76% of the total excess entropy at the highest density ($\rho^* = 0.70$) and over 90% at low to moderate densities ($\rho^* \leq 0.20$).

3. **Pair Entropy Capture of $\Delta s_{\mathrm{ex}}$**: The two-body difference $\Delta s^{(2)} = +0.0112\,k_B$ accounts for 58% of the total excess entropy difference $\Delta s_{\mathrm{ex}} = +0.0192\,k_B$, demonstrating that pair correlation functions capture the dominant structural differences between the two electrolyte solutions.

# 10 Scope of the Updated Transport Evidence

The historical HNC tables above concern $I = 0.40$ and 0.19, whereas the new replicated MD systems have $I = 0.12$ and 0.17. Their entropies and diffusivities cannot be mixed. The former transport comparison and its inferred $B \simeq 2.44$ have been replaced: a two-state ratio does not independently validate scaling, and the archived group-COM/species transport estimates were not consistently reconciled. The following sections use exclusively the completed eight-mixture study and share their numerical tables with the research manuscript.

For the new simulations the LJ potential is cut and shifted at $2.5\sigma$, with no tail correction, and the full Coulomb interaction has energy prefactor $\Gamma = 1$, hence $\lambda_B^* = \Gamma/T^* = 0.5$. All diameters and masses are one, and $\varepsilon_{ij} = \sqrt{\varepsilon_i \varepsilon_j}$. Excess entropy is measured relative to the ideal mixture at the same composition, including its ideal mixing contribution in the reference. Rosenfeld scaling uses $D_i^R = D_i \rho^{1/3} \sqrt{m_i/(k_B T)}$ [6, 7]. This bulk descriptor need not determine every species coefficient.

Table 6: Selected replicated MD systems. Every mixture contains $x_{+2} = x_{-2} = 0.03$ and has unit masses and nominal diameters. The primary neutral solvent and all ions have $\varepsilon = 1$. Fractions of secondary neutral solvents and their respective $\varepsilon$ values are listed in matching order; $x_{\pm 1}$ is the fraction of each monovalent sign. Label count and mechanically distinct species count are separated. $I = \frac{1}{2}\sum_i x_i q_i^2$.

| State | Labels/distinct | $x_{\varepsilon=1,\, q=0}$ | Secondary fractions | Secondary $\varepsilon$ | $x_{\pm 1}$ | $I$ |
|---|---|---|---|---|---|---|
| B3 | 3/3 | 0.94 | — | — | 0 | 0.12 |
| B5 | 5/5 | 0.84 | — | — | 0.05 | 0.17 |
| U4 | 4/4 | 0.64 | 0.30 | 0.25 | 0 | 0.12 |
| L5 | 5/4 | 0.64 | 0.15+0.15 | 0.25, 0.25 | 0 | 0.12 |
| M5 | 5/5 | 0.64 | 0.15+0.15 | 0.0625, 0.5625 | 0 | 0.12 |
| H5 | 5/5 | 0.64 | 0.15+0.15 | 0.01, 0.81 | 0 | 0.12 |
| U6 | 6/6 | 0.54 | 0.30 | 0.25 | 0.05 | 0.17 |
| H7 | 7/7 | 0.54 | 0.15+0.15 | 0.01, 0.81 | 0.05 | 0.17 |

# 11 HNC-Guided Replicated Molecular Dynamics Study

## 11.1 Composition Design and Controls

The new study tests whether adding physically distinct components can raise excess entropy and accelerate unchanged divalent probes. All selected mixtures have $x_{+2} = x_{-2} = 0.03$, equal masses and nominal diameters, $T^* = 2$, and target density $\rho^* = 0.7$. Table 6 defines the eight states. B3 and B5 reproduce the compositions of the original three-component and monovalent-background five-component baselines. The new solvent mixtures contain 30% secondary neutral material, rather than the 5% used in the earlier four-component dispersion comparison; they are distinct physical states.

Thirty-five integer-charge, mole-fraction, and dispersion candidates were screened with HNC on 2048 radial points with $\Delta r = 0.01\sigma$. All converged. The eight selected candidates were refined on 4096 points with $\Delta r = 0.005\sigma$. Their excess entropies agree with the coarse results at the reported $10^{-4}k_B$ precision. The first 20 sampled low-wavevector structure-factor matrices were positive for these states; this is a screening diagnostic, not a proof of phase stability. Charge-only candidates did not provide the desired large positive excess-entropy signal, so the replicated MD effort concentrates on solvent diversification and the original salt-addition comparison.

U4 contains one secondary neutral solvent with $\varepsilon = 0.25$. L5 splits that solvent into equal populations of mechanically identical labels. M5 and H5 instead split it into two physically distinct solvents with respective $\varepsilon$ pairs $(0.0625, 0.5625)$ and $(0.01, 0.81)$. Under geometric mixing, all four states have

$$\sum_i x_i\sqrt{\varepsilon_i} = 0.85, \qquad \sum_{ij} x_i x_j \varepsilon_{ij} = 0.85^2 = 0.7225. \tag{31}$$

Their ion fraction, ionic strength, nominal sizes, and masses are also identical. U6/H7 repeat the uniform/diverse comparison with 5% of each monovalent sign, retaining the same mean bare pair amplitude. Equation (31) does not fix actual potential energy, local solvation, effective excluded volume, or friction: the distribution of interactions still changes. H5/H7 are deliberately strong model perturbations, while M5 is a moderate counterpart.

## 11.2 Simulation and Entropy Integration

LAMMPS simulations [9] use $N = 4000$, including 120 probes of each divalent sign. Each composition has three independent replicas, with seeds 64123, 76237, and 91817. Within each replica index, compositions share initial particle positions and probe identities, enabling paired

comparisons. Species assignments are random on a perturbed cubic lattice, followed by energy minimization. All particles are integrated with a single Nosé–Hoover $NVT$ thermostat, damping time $0.1\tau$, and timestep $0.002\tau$.

Each replica covers 19 densities:

$$\begin{aligned}\rho^* = {} & 0.001, 0.0025, 0.005, 0.01, 0.02, 0.05, \\ & 0.10, 0.15, 0.20, 0.25, 0.30, 0.35, \\ & 0.40, 0.45, 0.50, 0.55, 0.60, 0.65, 0.70.\end{aligned} \tag{32}$$

Every state uses the same LJ potential, cut and shifted at $2.5\sigma$, with no tail correction. PPPM accuracy is $10^{-5}$; the real-space Coulomb cutoff is $5(0.7/\rho^*)^{1/3}\sigma$, below half the box length. Only the numerical electrostatic splitting changes with density. It must be distinguished from a change in the LJ Hamiltonian, which invalidated the original density-integration path. The Coulomb energy prefactor is one, giving $\lambda_B/\sigma = 0.5$ at $T^* = 2$.

The 18 lower-density stages use 50 000 equilibration and 100 000 production steps each. At $\rho^* = 0.7$, 100 000 equilibration steps precede 500 000 production steps ($1000\tau$); this transport run supplies the final thermodynamic point. All 24 allocations and 456 state stages completed and passed output verification.

Entropy integration uses the final half of production at each density. The default all-particle temperature has $3N-3$ kinetic degrees of freedom after removal of global momentum. In reduced units, the configurational pressure reference is

$$Z_{\mathrm{ref}} = 1 + \frac{\langle P\rangle - \rho(1-1/N)\langle T\rangle}{\rho T_0}, \qquad T_0 = 2. \tag{33}$$

Subtracting the sampled kinetic contribution and restoring the analytic ideal term removes its finite-$N$ offset and sampling noise from the excess EOS. It does not remove finite-size effects in configurational correlations.

For $a_{\mathrm{ex}} = A_{\mathrm{ex}}/(Nk_BT_0)$, define $y = \sqrt{\rho^*}$ and integrate

$$a_{\mathrm{ex}}(\rho^*) = \int_0^{\sqrt{\rho^*}} \frac{2[Z_{\mathrm{ref}}(y^2) - 1]}{y}\, dy. \tag{34}$$

The Debye–Hückel endpoint is finite in this variable:

$$F(0) = -\frac{[4\pi\lambda_B^* \sum_i x_i q_i^2]^{3/2}}{12\pi}. \tag{35}$$

A cubic spline through this endpoint and the simulated densities gives $a_{\mathrm{ex}}$, followed by $s_{\mathrm{ex}}/k_B = \langle U_{\mathrm{ex}}\rangle/(Nk_BT_0) - a_{\mathrm{ex}}$. Shape-preserving interpolation and alternative production-averaging windows test sensitivity. The ideal reference is always the ideal mixture at the same composition; its mixing entropy is excluded from $s_{\mathrm{ex}}$.

### 11.3 Transport Estimators and Replica Uncertainty

Unwrapped particle positions are recorded every $0.5\tau$ for the full dense production run. The same whole-system center-of-mass displacement is subtracted from every particle. Multiple time origins, approximately 200 per lag, provide species MSDs out to $250\tau$. A linear slope divided by six over $20$–$100\tau$ is primary; $10$–$50\tau$ and $50$–$200\tau$ are sensitivity windows. At every lag and for the fitted coefficients, $D_{\mathrm{net}} = \sum_i x_i D_i$ is checked. Since both probe populations are equal, $D_{\mathrm{probe}} = (D_{+2} + D_{-2})/2$; this is a mean self-diffusion coefficient, not diffusion of a neutral bound pair.

Velocities sampled every $0.01\tau$ during the final $100\tau$ permit species VACF/Green–Kubo checks, using the recorded whole-system COM velocity. All ionic velocities and up to 200 representative

Table 7: Dense-state HNC excess entropy and replicated MD means at $T^* = 2$, $\rho^* = 0.7$, $N = 4000$. Diffusion coefficients are in $\sigma^2/\tau$; $D_{\mathrm{probe}} = (D_{+2} + D_{-2})/2$. MD uses the primary 20–100$\tau$ MSD fitting window. L5 is an exact mechanical duplicate of U4 and is excluded from entropy-scaling fits. Confidence intervals for the primary comparisons are reported in Table 9; absolute-state intervals are in Table 8.

| State | HNC $s_{\mathrm{ex}}/k_B$ | MD $s_{\mathrm{ex}}/k_B$ | $D_{\mathrm{net}}$ | $D_{\mathrm{probe}}$ | $D_{+2}$ | $D_{-2}$ |
|---|---|---|---|---|---|---|
| B3 | -2.06350 | -1.97101 | 0.22253 | 0.20677 | 0.21196 | 0.20158 |
| B5 | -2.06770 | -1.97421 | 0.22279 | 0.21047 | 0.20879 | 0.21214 |
| U4 | -1.84340 | -1.77209 | 0.25945 | 0.23684 | 0.23913 | 0.23455 |
| L5 | -1.84340 | -1.77166 | 0.25945 | 0.23684 | 0.23913 | 0.23455 |
| M5 | -1.81020 | -1.74170 | 0.26720 | 0.24038 | 0.23937 | 0.24139 |
| H5 | -1.73270 | -1.66887 | 0.28401 | 0.24825 | 0.24764 | 0.24885 |
| U6 | -1.84780 | -1.77603 | 0.26018 | 0.23572 | 0.23742 | 0.23401 |
| H7 | -1.73700 | -1.67272 | 0.28196 | 0.24795 | 0.24634 | 0.24956 |

Table 8: Absolute-state nominal 95% Student-$t$ intervals across three independent replicas. Because the states are paired by replica, interval overlap here is not a test of the paired differences in Table 9. These intervals exclude systematic uncertainty.

| State | $s_{\mathrm{ex}}/k_B$ | $D_{\mathrm{net}}$ | $D_{\mathrm{probe}}$ |
|---|---|---|---|
| B3 | $[-1.97339, -1.96864]$ | $[0.21959, 0.22546]$ | $[0.20364, 0.20991]$ |
| B5 | $[-1.97657, -1.97186]$ | $[0.22095, 0.22463]$ | $[0.20626, 0.21468]$ |
| U4 | $[-1.77342, -1.77076]$ | $[0.25884, 0.26005]$ | $[0.23336, 0.24032]$ |
| L5 | $[-1.77421, -1.76912]$ | $[0.25884, 0.26005]$ | $[0.23336, 0.24032]$ |
| M5 | $[-1.74422, -1.73917]$ | $[0.26358, 0.27082]$ | $[0.22969, 0.25107]$ |
| H5 | $[-1.66936, -1.66838]$ | $[0.28214, 0.28588]$ | $[0.23921, 0.25728]$ |
| U6 | $[-1.77667, -1.77539]$ | $[0.25480, 0.26557]$ | $[0.21782, 0.25361]$ |
| H7 | $[-1.67584, -1.66960]$ | $[0.27882, 0.28511]$ | $[0.23406, 0.26183]$ |

particles of each neutral species are retained. Integrals at 5 and 10$\tau$ are diagnostics; the short velocity record does not establish a precise long-time plateau. Low-wavevector concentration fluctuations are monitored as a homogeneity diagnostic.

Uncertainty comes from independent replicas, not overlapping MSD origins. Absolute-state intervals use Student-$t$ statistics across three replicas. Composition effects use paired entropy differences and paired $\ln(D_a/D_b)$, with nominal 95% intervals and two degrees of freedom. Percent changes and interval endpoints are obtained by exponentiation. No multiple-comparison correction is applied; finite-size and estimator bias are outside these intervals. L5 is an exact mechanical duplicate of U4 and is never counted as another independent state in scaling fits.

# 12 Results and Discussion of the Replicated Study

## 12.1 Higher Excess Entropy Selected by HNC

Table 7 reports the new dense-state results; these replace the earlier single-trajectory transport estimates as the evidence for component-dependent mobility. The corresponding absolute-state intervals are given in Table 8. HNC successfully selects higher-excess-entropy liquids: H5 relative to U4 has $\Delta s_{\mathrm{ex}} = +0.1107 k_B$ in HNC and $+0.10322 k_B$ in MD. With monovalent background, H7 relative to U6 gives $+0.1108 k_B$ and $+0.10331 k_B$, respectively. Absolute HNC entropies remain more negative than MD by approximately 0.064–0.093$k_B$ per particle. Agreement of increments is better than agreement of absolute values.

Relative to their original baselines, H5 and H7 raise MD excess entropy by $0.30214 k_B$ and $0.30149 k_B$. The moderate split M5 raises it by $0.03039 k_B$ relative to U4. Entropy changes

Table 9: Paired composition changes, numerator relative to denominator. Brackets are nominal 95% Student-$t$ intervals from three independent replica pairs (two degrees of freedom), without multiple-comparison correction. Diffusion changes are $100[\exp(\overline{\ln(D_a/D_b)}) - 1]$ and intervals are transformed in the same way. Entropy differences are per particle in $k_B$. Intervals quantify replica sampling, excluding estimator and finite-size bias.

| Comparison | $\Delta s_{\mathrm{ex}}/k_B$ [95% CI] | $\Delta D_{\mathrm{net}}$ (%) [95% CI] | $\Delta D_{\mathrm{probe}}$ (%) [95% CI] |
|---|---|---|---|
| L5/U4 | $+0.00042$ $[-0.00219, +0.00304]$ | $+0.00$ $[+0.00, +0.00]$ | $+0.00$ $[+0.00, +0.00]$ |
| M5/U4 | $+0.03039$ $[+0.02727, +0.03352]$ | $+2.99$ $[+1.38, +4.62]$ | $+1.48$ $[-1.57, +4.64]$ |
| H5/U4 | $+0.10322$ $[+0.10209, +0.10434]$ | $+9.47$ $[+8.96, +9.97]$ | $+4.81$ $[+0.60, +9.19]$ |
| H5/B3 | $+0.30214$ $[+0.29927, +0.30501]$ | $+27.63$ $[+25.40, +29.90]$ | $+20.05$ $[+14.30, +26.09]$ |
| H7/U6 | $+0.10331$ $[+0.09990, +0.10672]$ | $+8.37$ $[+7.31, +9.44]$ | $+5.20$ $[-6.14, +17.92]$ |
| H7/B5 | $+0.30149$ $[+0.29823, +0.30476]$ | $+26.56$ $[+25.99, +27.13]$ | $+17.79$ $[+11.17, +24.81]$ |
| B5/B3 | $-0.00320$ $[-0.00543, -0.00097]$ | $+0.12$ $[-1.79, +2.06]$ | $+1.79$ $[-1.37, +5.04]$ |

induced by diversification are therefore substantially larger than the residual quadrature and averaging-window sensitivities assessed below.

### 12.2 Label Count, Ideal Mixing, and Physical Diversification

U4 and L5 provide an exact mechanical null. Splitting the secondary solvent into two identical labels increases formal ideal mixing entropy by $0.30 \ln 2 = 0.20794 k_B$ per particle if the labels are treated as distinguishable. Nevertheless, their diffusion coefficients are exactly identical in every replica. The measured excess-entropy difference is only $+0.00042 k_B$, with a nominal interval $[-0.00219, 0.00304] k_B$, consistent with thermodynamic sampling. Component labels alone do not supply a mechanical driving variable for diffusion.

M5 and H5 have the same mole fractions and ideal mixing entropy as L5, but different interactions. Thus the L5/M5/H5 entropy differences are interaction-induced excess changes, not additional ideal mixing. Adding monovalent salt to the baseline, B3→B5, increases the ideal mixing contribution by $0.38787 k_B$ but slightly *lowers* excess entropy: $\Delta s_{\mathrm{ex}} = -0.00320 k_B$, interval $[-0.00543, -0.00097] k_B$. Neither mean diffusion ($+0.12\%$) nor probe diffusion ($+1.79\%$) has a resolved change. More charged components therefore do not by themselves guarantee higher excess entropy or faster probes.

### 12.3 Divalent-Ion Enhancement: Large and Controlled Changes

The largest design changes produce clear increases in divalent-ion mobility. H5/B3 gives $+20.05\%$, interval $[14.30, 26.09]\%$, while H7/B5 gives $+17.79\%$, interval $[11.17, 24.81]\%$. Both signs independently increase at the primary window: $+16.8\%$ for $+2$ and $+23.4\%$ for $-2$ in H5/B3, and $+18.0\%$ and $+17.6\%$ in H7/B5. All four separate-sign paired nominal intervals exclude zero. These comparisons also alter mean interaction strength and local friction, so entropy is a correlate rather than an isolated causal variable.

The more controlled diversification comparison gives a smaller response. U4→H5 preserves charged composition, ionic strength, masses, nominal diameters, and mean bare pair amplitude. Mean diffusion increases by $9.47\%$, interval $[8.96, 9.97]\%$, whereas probe diffusion increases by $4.81\%$, interval $[0.60, 9.19]\%$. With monovalent background, U6→H7 increases mean diffusion by $8.37\%$, interval $[7.31, 9.44]\%$, and probe diffusion by $5.20\%$, interval $[-6.14, 17.92]\%$. The latter ion effect is unresolved. The moderate split gives a clear mean increase of $2.99\%$, but its $1.48\%$ probe increase is also unresolved. The ions respond less strongly than the solvent-dominated mean.

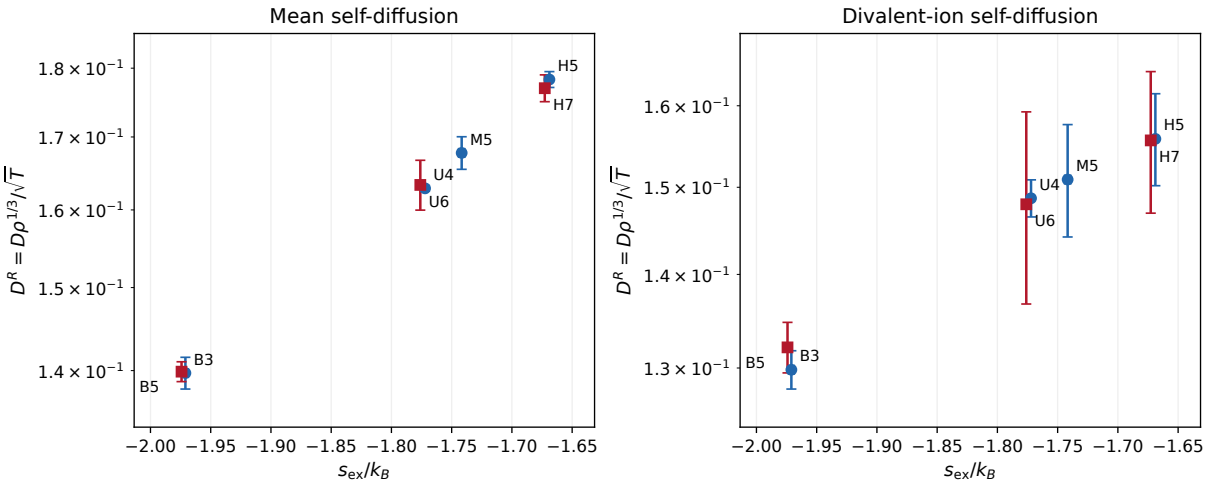



Figure 1: Rosenfeld-reduced mean and divalent-probe self-diffusion versus replicated MD excess entropy. Vertical axes are logarithmic. Blue circles: no monovalent background; red squares: 5% of each monovalent sign. Bars are nominal 95% absolute-state replica intervals. L5 coincides exactly with U4 in transport and is omitted as a duplicate. The broad positive trend does not fix the slope of controlled composition changes.

### 12.4 Rosenfeld-Like Correlation and Its Limits

Figure 1 shows a positive entropy–mobility trend for both the mean and divalent probes. All masses, temperatures and densities are equal, so $D_i^R = D_i\rho^{1/3}/\sqrt{T}$ differs from $D_i$ by the same constant; diffusion ratios are unchanged. Descriptive unweighted fits of composition means to

$$\ln D_i^R = \ln A_i + B_i s_{\mathrm{ex}}/k_B \tag{36}$$

give $B_{\mathrm{net}} = 0.80$ and $B_{\mathrm{probe}} = 0.62$ for B3/U4/M5/H5, and 0.78 and 0.55 for B5/U6/H7. All four fits have $R^2 > 0.99$. These fits contain only four or three composition means, exclude the duplicate L5 state, and do not incorporate uncertainty in both coordinates; the high $R^2$ is descriptive rather than an independent validation of a universal law.

The controlled local response differs from the broad-family fit. For H5/U4, $\Delta \ln D/\Delta(s_{\mathrm{ex}}/k_B)$ is approximately 0.88 for the mean and 0.46 for probes; H7/U6 gives approximately 0.78 and 0.49. The probe response is weaker than the mean response under the same intervention. Mean self-diffusion assigns only 6% weight to divalent particles and cannot determine their mobility separately. The new results support a restricted entropy–mobility association for both observables, while neither component count nor a single bulk entropy establishes a universal species law. No association-lifetime mechanism has yet been measured in this study.

### 12.5 Estimator Sensitivity and Charge-Sign Symmetry

The controlled ion point estimates remain positive, approximately 4–6%, across the MSD windows (Table 10 and Fig. 2). However, H5/U4 loses nominal significance over 50–200$\tau$: +3.94%, interval $[-8.44, 18.00]$%. H7/U6 is unresolved in all three windows. Accordingly the approximately 5% ion effect is a provisional trend, not a converged precise enhancement. VACF integrals support the broad mobility ordering, but do not independently resolve such a small effect with the present final-100$\tau$ records.

Charge-sign symmetry remains a convergence diagnostic. The B3 model is invariant under global charge reversal, yet its primary-window estimate gives $D_{+2} - D_{-2} = 0.01039$, nominal replica interval $[0.00291, 0.01786]$. The other states have sign-difference intervals containing zero. Generated inputs use identical masses and LJ parameters for both probe signs, random probe

Table 10: Sensitivity of the paired divalent-ion diffusion change (percent) to MSD fitting window, with nominal 95% replica intervals. The 20–100$\tau$ window is primary. The positive controlled effect is not uniformly significant across estimators/windows.

| Comparison | 10–50$\tau$ | 20–100$\tau$ | 50–200$\tau$ |
|---|---|---|---|
| H5/U4 | +5.92 [+1.87, +10.13] | +4.81 [+0.60, +9.19] | +3.94 [−8.44, +18.00] |
| H7/U6 | +5.31 [−1.17, +12.22] | +5.20 [−6.14, +17.92] | +4.46 [−5.66, +15.67] |

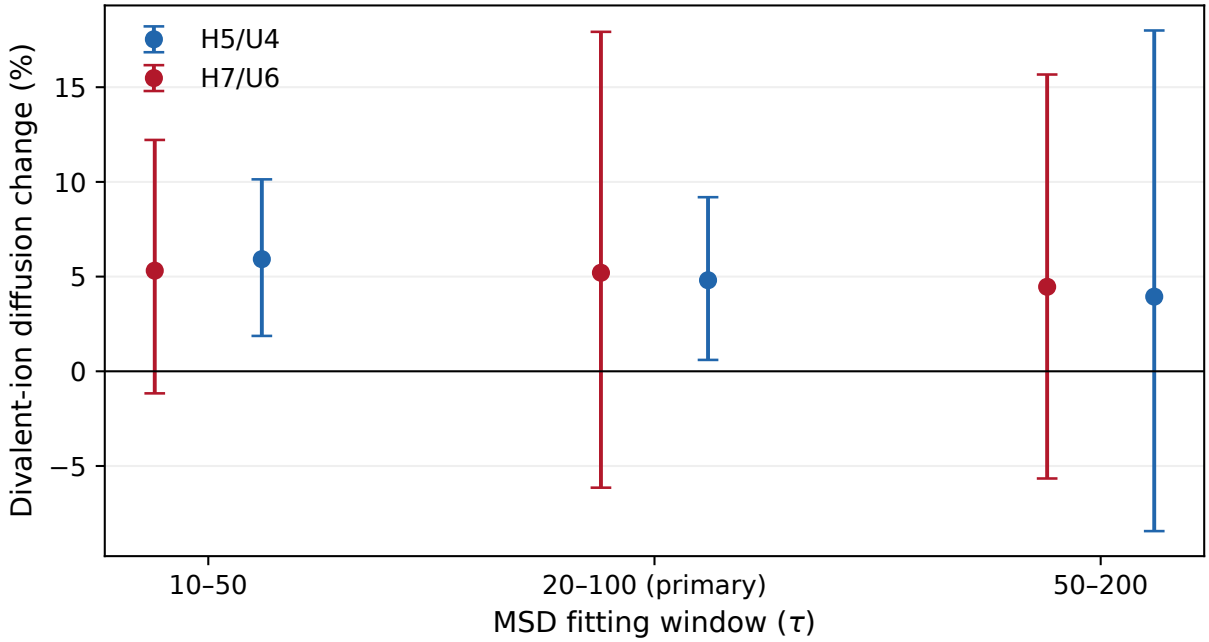



Figure 2: Controlled divalent-ion diffusion changes and nominal paired 95% intervals for alternative MSD windows. The point estimates are consistently positive, but statistical significance is sensitive to the fitting window and monovalent background.

assignments, one all-particle thermostat, and a common COM frame; no sign-dependent input parameter was found. The B3 asymmetry nevertheless flags remaining sampling or uncertainty-estimation limitations. Pooling the signs does not remove that concern. It also does not explain away the large baseline contrasts, for which each sign separately increases.

Across all 24 replicas, switching from cubic to shape-preserving interpolation changes dense-state entropy by at most $0.00032k_B$. Retaining all or 75% of production instead of the final half changes it by at most $0.00165k_B$. These sensitivities are small relative to the $0.103k_B$ diversification increments, but they do not bound finite-size bias or error below the lowest simulated density. Low-wavevector species concentration-fluctuation diagnostics remain of order unity, approximately 0.48–1.80 for individual replica/species values, with no large enhancement in the selected mixtures. One box size and these trajectory lengths cannot exclude slower segregation or establish phase stability.

### 12.6 Rehabilitation and Limits of the 2PT Excess Entropy

We reanalyzed the existing final-100$\tau$ velocity records of all 24 replicas using a corrected classical two-phase thermodynamic (2PT) estimator. The revised spectral construction follows Sun *et al.* [10], with the equal-partial-volume mixture approximation discussed by Lai *et al.* [11]. Each species is assigned $\bar{V}_i = 1/\rho$; equal nominal diameters do not establish equality of actual partial volumes in these charged, dispersion-diverse liquids. The primary gas/solid partition exponent

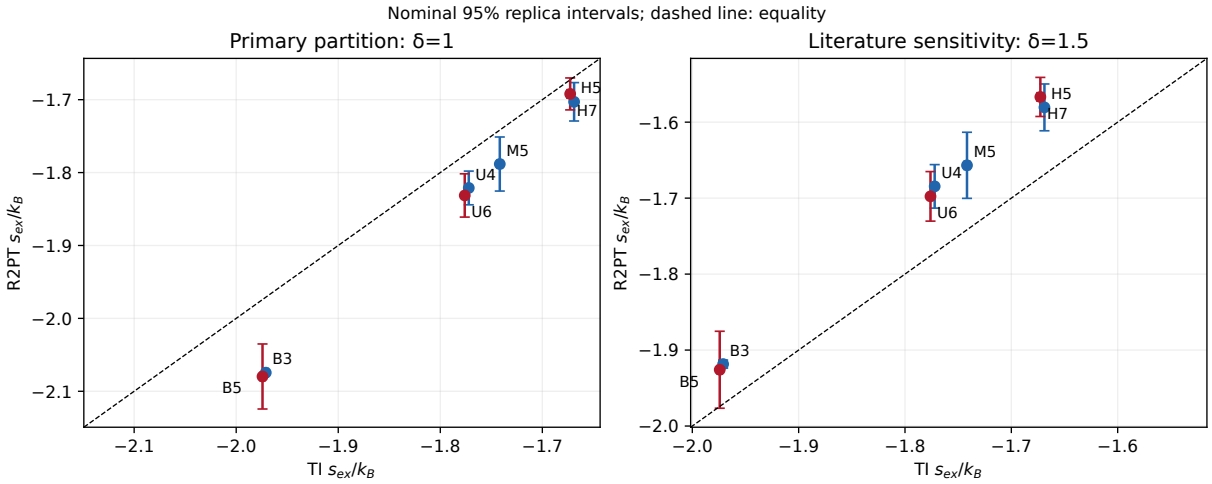



Figure 3: Corrected classical revised-2PT excess entropy against TI for the seven mechanically distinct mixtures. L5 duplicates U4 in transport and is omitted. Bars are nominal 95% absolute-state replica intervals; dashed lines denote equality. The left panel uses the primary partition $\delta = 1$, and the right panel the unfitted literature sensitivity $\delta = 1.5$. Internal consistency does not remove composition-dependent thermodynamic bias.

is $\delta = 1$; the published LJ value $\delta = 1.5$ is an unfitted sensitivity. Neither partition, reference volume, nor entropy offset is adjusted to match thermodynamic integration (TI).

The calculation corrects the density scaling and prefactor of the normalized diffusivity ($\Delta \propto D\sqrt{m/T}\rho^{1/3}$), weights gas entropy by its gas-like fraction, and subtracts the classical ideal-mixture reference consistently. The complete finite-record autocorrelation, in the common system-COM frame, is Gaussian-tapered with primary width $10\tau$. Its spectrum spans the full Nyquist band and is normalized by measured $C(0)$ to three modes per particle; raw target-temperature mode counts are retained as a diagnostic. The gas Lorentzian and its entropy tail are integrated analytically to infinity. The residual spectrum is kept signed as revised-2PT bookkeeping, rather than clipped to a positive "solid population." This partition is not a literal count of free ions or bound oscillators.

Seven analytic checks pass, including the standard fluidicity polynomial, unit invariance, a solvable VACF/Green–Kubo limit, the Lorentzian entropy tail, the dilute-gas limit, and classical-reference and label/mixing cancellation. Normalization, reconstruction, and independence of the arbitrary classical phase-space constant hold to floating-point precision. Mechanically identical labels are pooled before the nonlinear entropy estimate. For all three actual U4/L5 pairs, matched retained particle IDs have identical COM-corrected velocity hashes, VACFs, and resulting 2PT entropies. Separately processing the label samples gives a difference $+0.00062k_B$, interval $[-0.00428, 0.00553]k_B$, consistent with sampling. State-mean pooled-probe spectral diffusivities agree with the full-production MSD coefficients within 2.6%.

These successes rehabilitate the implementation, but not its quantitative thermodynamic accuracy across compositions (Fig. 3). For B3 the primary estimate is $-2.07445k_B$, compared with TI $-1.97101k_B$: the approximately $3.4k_B$ discrepancy of the archived $-5.376k_B$ result is reduced to $0.10344k_B$. Primary absolute biases range from about $-0.019$ to $-0.105k_B$, with all nominal paired bias intervals below zero. B3, B5, and U6 exceed the prespecified practical mean-bias tolerance $0.05k_B$. Thus the remaining error is composition dependent and comparable to the controlled entropy signal; a constant offset is insufficient.

For H5/U4, primary 2PT gives $\Delta s_{\mathrm{ex}} = 0.11818k_B$ versus TI $0.10322k_B$, a point bias $+0.01496k_B$ (Table 11). This meets the practical point-bias tolerance $0.02k_B$ and is statistically consistent with zero bias, but the wide interval does not establish equivalence within that tolerance. For H7/U6, the estimate is $0.13927k_B$ versus $0.10331k_B$, an approximately 35%

Table 11: Controlled excess-entropy increments from TI and corrected classical revised 2PT. All entropy quantities are per particle in $k_B$. Bias is the paired difference between the 2PT and TI increments; brackets are nominal 95% Student-$t$ intervals from three independent replica pairs (two degrees of freedom), without multiplicity correction. The primary partition is $\delta = 1$; $\delta = 1.5$ is an unfitted sensitivity.

| Comparison | Partition | TI $\Delta s_{\mathrm{ex}}/k_B$ | 2PT $\Delta s_{\mathrm{ex}}/k_B$ | Bias [95% CI] |
|---|---|---|---|---|
| H5/U4 | $\delta = 1$ | 0.10322 | 0.11818 | $+0.01496$ $[-0.00544, +0.03536]$ |
| H5/U4 | $\delta = 1.5$ | 0.10322 | 0.10397 | $+0.00075$ $[-0.02175, +0.02326]$ |
| H7/U6 | $\delta = 1$ | 0.10331 | 0.13927 | $+0.03596$ $[+0.01501, +0.05692]$ |
| H7/U6 | $\delta = 1.5$ | 0.10331 | 0.13086 | $+0.02755$ $[+0.00705, +0.04805]$ |

overestimate; its bias interval $[0.01501, 0.05692]k_B$ excludes zero. The $\delta = 1.5$ partition improves the first increment to $0.10397k_B$ but still overestimates the second at $0.13086k_B$. Agreement for one pair does not validate a partition for all compositions.

The discrepancies survive sensitivity checks. State-mean entropy spans over taper widths 5, 10, and $20\tau$ are 0.011–$0.022k_B$; first/last-half record shifts reach $0.012k_B$. Frequency-grid refinement changes means by less than $0.00002k_B$, and sampling-interval decimation changes them by at most $0.0022k_B$. Illustrative volume-conserving $\pm 10\%$ partial-volume perturbations shift means by at most $0.0032k_B$, without establishing that the equal-volume approximation is exact. Positive-part clipping would add modes: the signed negative residual represents approximately 2.3–3.9% of the total mode weight in primary species spectra. The corrected bookkeeping avoids this artifact, but cannot eliminate the model dependence of the gas/solid partition.

We therefore retain TI as the entropy coordinate in the Rosenfeld analysis. Corrected 2PT and the underlying VACFs remain useful dynamical diagnostics, but species spectral allocations are not independently measured partial molar entropies. Because diffusion enters the zero-frequency spectrum and fluidicity construction, a 2PT–diffusion correlation is not an independent confirmation of entropy-driven enhancement. Probe fluidicity rises from about 0.530 in B3 to 0.569 in H5, consistent with increased low-frequency motion. However, the mean probe VACFs remain positive over the first $\tau$; a pronounced early negative backscattering minimum is not supported. Association populations and residence times, rather than a literal interpretation of the 2PT partition, are needed to test local-caging mechanisms.

### 12.7 Further Discriminating Checks

The first priority is longer independent matched replicas of U4/H5 and U6/H7, together with extended B3 sampling to resolve the charge-sign flag. Next, repeat matched states with $N = 8000$ or 16000 at fixed density and composition. Box-size effects need not cancel between liquids with different viscosity. Particle-size variation is a separate perturbation of packing, contact electrostatic coupling, and friction and should not be conflated with this box-size test.

Existing trajectories permit probe–solvent and probe–ion RDFs, coordination populations, association lifetimes, and exchange-time analysis. Species-resolved pair entropy could help explain local mobility, but is not a thermodynamic partial molar entropy or a replacement for total excess entropy. These measurements would test local caging and association mechanisms without inferring kinetics from static entropy alone.

After convergence tests, increasing $x_{+2} = x_{-2}$ symmetrically, for example to 0.06 or 0.10, would strengthen the ionic contribution and alter screening and association. Uniform/diverse solvents should be compared at matched concentration and ionic strength within each pair, with HNC re-screening and homogeneity checks. Several densities and temperatures at fixed composition, and states with matched excess entropy but different compositions, would distinguish state-point scaling from composition-induced correlations.

## 13 Summary and Conclusions

The theoretical framework separates absolute classical entropy, total thermodynamic excess entropy, and its pair-correlation approximation. Internal HNC free-energy/virial agreement is a solver consistency check; it does not imply agreement with MD or validate a transport law. Historical pair-entropy percentages and compositions are not transferred to the new eight-state data. The correct screened endpoint also replaces the historical finite-intercept density extrapolation.

The replicated calculations establish a positive excess-entropy–mobility association in the selected solvent-design families, with clear limits on its interpretation.

1. **Component count alone has no demonstrated transport effect.** An exact label-only split raises formal ideal mixing entropy while leaving diffusion identical. Monovalent salt addition slightly lowers excess entropy and gives no resolved divalent speedup.

2. **HNC-guided interaction changes produce higher-excess-entropy, faster-ion liquids.** MD confirms entropy increases of approximately $0.30k_B$ and divalent diffusion gains of 18–20% relative to the original baselines. Both signs separately increase. These are successful design comparisons, not isolated entropy interventions.

3. **Controlled diversification gives a weaker, provisional ion response.** At fixed charged composition and mean bare pair amplitude, excess entropy rises by approximately $0.103k_B$, mean diffusion by 8–9%, and probe diffusion by approximately 5%. The ion significance is sensitive to fitting window and unresolved with monovalent background. Longer independent sampling and the B3 charge-sign convergence check remain necessary.

4. **Rosenfeld-like trends extend to probes but do not establish a universal law.** Broad-family fits have positive slopes and high descriptive $R^2$, with weaker slopes for probes than for the mean. Three or four composition means cannot determine universality, and changing interactions also changes local friction and solvation. Species-resolved kinetics, box-size comparisons, and state-point validation are the next discriminating tests.

5. **Corrected 2PT is internally consistent but not a quantitative TI replacement.** All 24 records pass analytic/reference and mechanical-label checks, while state-mean probe spectral diffusion agrees with MSD within 2.6%. Entropy biases nevertheless vary with composition: the controlled monovalent-background increment is overestimated by approximately 35% in the primary model. The partition sensitivity does not consistently remove this error. We retain TI for the entropy–mobility comparison and use spectra as supporting dynamics, not an independent scaling validation or a literal free-ion fraction.

## References


[1] A. R. Allnatt, *Renormalization of the Ornstein-Zernike Equation for Ionic Solutions*, Mol. Phys. **8**, 533 (1964).

[2] T. Morita and K. Hiroike, *A New Approach to the Theory of Classical Fluids. III*, Prog. Theor. Phys. **25**, 537 (1961).

[3] S. J. Singer and D. Chandler, *Free-Energy Functions in the Extended RISM Approximation*, Mol. Phys. **55**, 621 (1985).

[4] E. Lomba, *An Efficient Procedure for Solving the Reference Hypernetted Chain Equation (RHNC) for Simple Fluids*, Mol. Phys. **68**, 87 (1989).

[5] D. C. Wallace, *On the Interpretation of Liquid-State Entropy*, J. Chem. Phys. **87**, 2282 (1987).

[6] Y. Rosenfeld, *Relation Between the Transport Coefficients and the Internal Entropy of Simple Systems*, Phys. Rev. A **15**, 2545 (1977).

[7] Y. Rosenfeld, *A Quasi-Universal Excess-Entropy Scaling of Transport Properties in Simple and Complex Fluids*, J. Phys.: Condens. Matter **11**, 5415 (1999).

[8] M. Dzugutov, *A Universal Scaling Law for Atomic Diffusion in Condensed Matter*, Nature **381**, 137 (1996).

[9] S. Plimpton, *Fast Parallel Algorithms for Short-Range Molecular Dynamics*, J. Comput. Phys. **117**, 1 (1995).

[10] T. Sun, J. Xian, H. Zhang, Z. Zhang, and Y. Zhang, J. Chem. Phys. **147**, 194505 (2017), doi:10.1063/1.5001798.

[11] P.-K. Lai, C.-M. Hsieh, and S.-T. Lin, Phys. Chem. Chem. Phys. **14**, 15206 (2012), doi:10.1039/C2CP42011B.